

Barcelona Airbnb Market Intelligence Report

Expernetic Data Engineering Intern — Technical Assignment

Prepared by: [Your Name] **Date:** July 2026 **Dataset:** Inside Airbnb — Barcelona, June 2026 Snapshot **City analyzed:** Barcelona, Spain (single city, depth-first approach)

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1. Executive Summary

Barcelona's Airbnb market (15,293 listings, June 2026) is a mature, commercially-dominated short-term rental market operating under an accelerating regulatory crisis. Despite surface-level diversity — 4,595 unique hosts, 69 neighbourhoods, price range €11–€4,463/night — market supply is heavily concentrated: the top 10.9% of hosts control 61.1% of all listings, and a single operator manages 588 properties (3.8% of the entire city market).

The market's structural and regulatory risk is significant. Only 890 verified HUT tourist licenses were identified across 15,293 listings, while all ~10,101 existing city-wide licenses expire in November 2028 and will not be renewed — a Constitutional Court-upheld policy that effectively phases out Barcelona's entire short-term rental apartment market within 28 months of this analysis.

Analytically, this study produced five statistically confirmed hypotheses, a machine learning price model (XGBoost, $R^2=0.857$, $MAE=€50.36$), VADER sentiment analysis on 5,885 English reviews, and LLM-generated market intelligence briefings. Key actionable findings: entire-home listings command a 161% price premium over private rooms (large effect, $p<0.001$); Eixample is the sole standalone premium neighbourhood tier; superhost status drives 3x booking volume at equivalent pricing; and minimum stay length — not location — is the single strongest price predictor, indicating that the short-stay and medium-term rental markets coexist in this dataset and should be modeled separately.

2. Objectives & Scope

2.1 Assignment Objectives

This assignment required transforming raw Inside Airbnb public data into engineering artifacts, analytical insights, and business recommendations across five domains: data engineering, EDA, statistical analysis, machine learning, and applied AI/NLP.

2.2 City Selection Rationale

Barcelona was selected as the single analysis city for the following reasons:

- Rich, well-populated dataset (15,293 listings) with sufficient statistical power for hypothesis testing and ML modeling
- Strong external context (2028 regulatory phase-out) enabling business narrative beyond pure data description
- Mid-size complexity: large enough for meaningful analysis, small enough for depth-first treatment within the assignment timeline
- Less overused than NYC/London in data portfolios

2.3 Prioritization Rationale

Per the assignment's explicit design philosophy ("quality outweighs quantity"), this submission prioritizes:

- Exceptional depth in Sections 2-7 over superficial coverage of all sections
- Rigorous assumption-checking in statistical analysis over mechanical test-running
- Honest documentation of failures, corrections, and limitations over presenting only clean results
- Real, verified business context (regulatory research) over generic "business implications"

2.4 Sections Completed

[See README for full completion table]

3. Dataset Overview

3.1 Source

Inside Airbnb (insideairbnb.com), Barcelona snapshot scraped June 2026. An independent, non-commercial project that scrapes publicly available Airbnb listing data.

3.2 Files Used

File	Rows	Columns	Notes
listings.csv.gz (detailed)	15,293	90	Primary listings data
calendar.csv.gz	5,623,190	5	Availability only, no price
reviews.csv.gz (detailed)	1,033,523	6	Full review text
neighbourhoods.csv	75	2	Boundary names

File	Rows	Columns	Notes
neighbourhoods.geojson	75 features	—	Polygon boundaries

3.3 Entity Relationships

- **Listings** (grain: 1 row/listing) — core entity
- **Calendar** (grain: 1 row/listing/date) — 365 days forward per listing
- **Reviews** (grain: 1 row/review) — historical guest reviews
- **Hosts** (derived from listings) — 4,595 unique hosts
- **Neighbourhoods** — 69 with listings, 75 boundary polygons

3.4 Key Limitations

1. **12 columns 100% null** in listings (host_since, host_response_time, instant_bookable, etc.) — scrape-level gap, not a local error
2. **Calendar lacks pricing** — only availability + min/max nights; revenue estimation relies on pre-computed `estimated_revenue_1365d`
3. **Review text is multilingual** — 41.1% non-English; VADER sentiment analysis limited to English subset
4. **Single snapshot** — no longitudinal data; temporal trends inferred from review dates and calendar forward-bookings only
5. **Self-reported fields** — license, property type, amenities are host-declared and unverified

3.5 Assumptions

[All assumptions documented in reports/assumptions_and_decisions_log.md]

4. Methodology

4.1 Overall Approach

A depth-first, single-city approach was chosen deliberately. The analytical pipeline followed this sequence:

1. Profile raw data before writing any cleaning logic
2. Document every finding and decision before proceeding
3. Verify hypotheses about data quality programmatically, not visually
4. Apply demand segmentation early (active/dormant/never-active) so all downstream analysis uses appropriate population filters
5. Check statistical assumptions before selecting tests
6. Report effect sizes alongside p-values throughout

4.2 Key Method Selections

Decision	Choice	Rationale
Storage format	DuckDB + Parquet	Column-oriented, fast for analytics, no server required

Decision	Choice	Rationale
Statistical tests	Mann-Whitney U / Kruskal-Wallis	Price/rating data severely non-normal; non-parametric required
ML target	log(price)	Reduces right-skew; improves linear model performance
Sentiment tool	VADER	Designed for short informal text; no GPU required at this scale
LLM provider	Groq (Llama-3.3-70b)	Free tier, fast inference, sufficient for structured summarization

5. Engineering Approach

5.1 Pipeline Architecture

data/raw/ (gz files) ↓ src/profile_data.py Raw profiling report ↓ src/clean_listings.py ↓ src/clean_calendar.py ↓ src/clean_reviews.py data/processed/ (parquet files) ↓ src/build_warehouse.py data/warehouse.duckdb (star schema) ↓ notebooks/ EDA → Stats → ML → NLP ↓ reports/ (figures, logs, briefings, PDF)

5.2 Star Schema Design

Dimensions:

- `dim_host` — 4,595 rows (grain: host)
- `dim_listing` — 15,293 rows (grain: listing, descriptive attributes)
- `dim_neighbourhood` — 69 rows
- `dim_date` — 374 rows (generated from calendar range)

Facts:

- `fact_listing_performance` — 15,293 rows (grain: listing, numeric KPIs)
- `fact_calendar` — 5,581,945 rows (grain: listing × date)
- `fact_reviews` — 1,026,820 rows (grain: review)

View:

- `v_listing_demand` — demand segmentation (active/dormant/never_active)

5.3 Data Quality Results

Dataset	Raw Rows	Issues Found	Action
Listings	15,293	12 cols 100% null, 12.7% price null	Dropped dead cols, flagged
Calendar	5,623,190	No price column, 113 orphan IDs	Documented, LEFT JOIN
Reviews	1,033,523	114 null text, 77 orphan IDs	Dropped nulls, LEFT JOIN

5.4 Engineering Decision Log

Full decision log: [reports/assumptions_and_decisions_log.md](#) Key decisions: Parquet over CSV, LEFT JOIN orphan strategy, demand segmentation as warehouse view, log-transform of price target for ML.

5.5 Production Readiness Discussion

For production deployment this pipeline would require:

- **Orchestration:** Prefect or Airflow DAG replacing manual script execution
- **Incremental processing:** CDC on listings table (track scrape_id changes)
- **Monitoring:** Row-count and null-rate alerts on each pipeline stage
- **Containerization:** Docker compose for environment reproducibility
- **Cloud:** S3 (raw) → Glue (transform) → Athena/Redshift (warehouse)
- **Multi-city scaling:** Config-driven city parameter; parallel processing via Prefect map() or Spark for 50+ cities

6. EDA Findings

6.1 Price Distribution

- Active listing median: €213.68 (vs full dataset €177.67 — dormant listings skew cheap, not expensive)
- Strong right-skew (skewness 5.42 for entire homes)
- Entire homes: median €240.10 | Private rooms: median €92.00
- 36.8% of listings have zero trailing-12-month occupancy

Business interpretation: The "typical" Barcelona Airbnb costs €213.68/ night for an active listing — 20% higher than naive statistics suggest, because inactive/dormant listings disproportionately skew cheap. Benchmarking against the raw mean (€240.62) or median (€177.67) without demand-filtering would either over- or understate competitive pricing.

6.2 Geographic Analysis

Three clear pricing tiers by neighbourhood group (all differences statistically confirmed in Section 7 / H4):

- **Premium:** Eixample (€257.50) — standalone tier
- **Upper-mid:** Sant Martí (€230), Gràcia (€217)
- **Mid:** Les Corts (€193), Sants-Montjuïc (€201)
- **Budget:** Nou Barris (€80), Sant Andreu (€92), Ciutat Vella (€125)

Notable: Barceloneta (famous beachfront) sits in the budget tier at €88.86 median — a bimodal market of budget tourist flats and premium beachfront properties, with the median reflecting the larger budget segment rather than the neighbourhood's reputation.

6.3 Temporal & Seasonal Trends

- Historical demand (reviews) peaks in **May**
- Forward booking pressure (calendar availability) tightest in **July**
- Divergence explained by booking lead-time, not contradiction
- Winter trough: Dec-Jan (~35% below peak)

- Recency artifact: 2026-06/07 review data truncated to avoid misrepresenting the review-lag as a real demand drop

6.4 Host & Supply-Side Analysis

- Top 500 hosts (10.9%) control 61.1% of all listings
- Single largest operator: 588 listings (3.8% of market)
- 65% of hosts are single-listing casual operators — but control only 19.5% of inventory
- Superhost listings: 25.9% of market, 3x more reviews, same price

6.5 Review & Demand-Side Analysis

- Price ↔ rating correlation: $r=0.073$ (essentially zero)
- High-review-count + low-rating: only 3 listings total (near-absent pattern)
- 2 of those 3 share the same host — not an independent market segment

7. Statistical Findings

All tests used non-parametric methods (Mann-Whitney U / Kruskal-Wallis) due to confirmed violations of normality (skewness 2.53–8.97) and homoscedasticity (Levene's $p<0.000001$) in all tested variables.

Hypothesis	Test	Result	Effect Size	Conclusion
H1: Entire home > private room price	Mann-Whitney U	$p<0.001$	$r=0.695$ (large)	✓ Confirmed
H2: Superhost rating > non-superhost	Mann-Whitney U	$p<0.001$	$r=0.438$ (moderate-large)	✓ Confirmed
H3: 10+ reviews ≠ fewer reviews (price)	Mann-Whitney U	$p<0.001$	$r=0.544$ (large)	✓ Confirmed
H4: Neighbourhood prices differ	Kruskal-Wallis + Dunn's	$p<0.001$	$\eta^2=0.095$	✓ Confirmed
H5: Weekend ≠ weekday availability	Chi-square	$p<0.001$	$V=0.0035$ (negligible)	⚠ Stat. sig, not practical

Key insight: H5 demonstrates that statistical significance without effect size is misleading — with 5.58M observations, a 0.39pp difference tests as "significant" but is operationally meaningless. This is why effect size reporting is non-negotiable.

H4 detail: 33 of 45 Bonferroni-corrected pairwise comparisons are significant. Non-significant pairs reveal the three-tier structure. Neighbourhood group explains 9.5% of price variance ($\eta^2=0.095$) — real but not the dominant driver (room type from H1 explains more).

8. Data Science Experiments

8.1 Feature Engineering

27 features engineered from raw data including:

- Physical attributes: accommodates, bathrooms, bedrooms, beds, beds_per_person
- Listing type: room_type (ordinally encoded), minimum_nights
- Performance: number_of_reviews, review_scores (4 dimensions), estimated_occupancy_l365d, availability_365
- Host: host_listings_count, hosts_time_as_host_years, is_superhost
- Geography: 9 neighbourhood group dummies (Eixample as reference dropped)
- Engineered flags: has_rating, beds_per_person

Target: log1p(price) — log transform reduces right-skew, improves linear model calibration, and allows inverse-transform for interpretable MAE/RMSE reporting.

8.2 Model Comparison (5-fold CV)

Model	R ²	MAE (EUR)	RMSE (EUR)	Train R ²	Overfit
Ridge Regression	0.805	€57.13	€115.50	0.808	0.003
Random Forest	0.845	€52.48	€110.64	0.913	0.068
XGBoost	0.857	€50.36	€106.86	0.940	0.083

XGBoost selected as final model. MAE of €50.36 on median price of €213.68 = ~23.6% median error — respectable for unstructured real-world data without text/amenity features.

8.3 SHAP Feature Importance

Top 5 by mean |SHAP value|:

1. **minimum_nights** (0.378) — dominant predictor; acts as short-stay vs medium-term rental proxy, not a genuine "minimum stay preference" signal
2. **room_type_enc** (0.171) — consistent with H1's 161% price gap finding
3. **accommodates** (0.138) — physical capacity outweighs location
4. **bathrooms** (0.050)
5. **number_of_reviews** (0.045)

ng_Eixample (0.025) is the only neighbourhood dummy with meaningful weight — corroborates H4's finding that Eixample is the sole standalone premium tier.

8.4 Residual Analysis

- Entire home / Private room: near-zero median error (well-calibrated for 95% of data)
- Shared rooms: -10.5% median error (systematic overprediction — thin segment with insufficient training signal)
- Nou Barris: -15.5% mean error (model anchors too high for cheapest district — training signal dominated by pricier central listings)
- Horta-Guinardó: std=50.47% (highest variance — small-n heterogeneous district)

9. AI/ML Experiments

9.1 Sentiment Analysis (VADER)

- 10,000 reviews sampled, language-detected via langdetect
- 58.9% English (5,885 reviews) used for VADER analysis
- Non-English correctly excluded (VADER is English-only; multilingual reviews artifactually score as neutral)

Correlations (VADER compound vs numerical scores):

Score dimension	r
Overall rating	0.245
Value	0.236
Cleanliness	0.209
Location	0.117
Comment length	0.071

Location shows the weakest correlation — guests describe location objectively ("close to metro") rather than emotionally, so sentiment tools miss this dimension. Sentiment and numerical scores are complementary signals, not substitutes.

9.2 LLM Market Intelligence (Groq / Llama-3.3-70b)

A structured market intelligence briefing generator was built, taking verified analytical findings as grounded context and generating:

- Executive briefing (C-suite / investor audience)
- Host briefing (individual property operator audience)

Key design choices to minimize hallucination:

- `temperature=0.3` (low, favors factual output)
- Explicit instruction: "only reference findings explicitly stated above"
- All statistics pre-computed from data, passed as context — LLM acts as writer, not analyst
- Output saved to `reports/market_intelligence_briefings.txt`

Full briefing text available in reports directory.

10. Visualizations

Figure	File	Section
Price distribution + by room type	price_distribution.png	6.1
Neighbourhood price choropleth	price_choropleth.png	6.2
Seasonality dual-axis (reviews + availability)	seasonality_dual.png	6.3
Neighbourhood price tiers bar chart	neighbourhood_price_tiers.png	6.4

Figure	File	Section
SHAP feature importance (bar)	shap_importance.png	8.3
SHAP beeswarm	shap_beeswarm.png	8.3
Residual analysis (4-panel)	residual_analysis.png	8.4
Sentiment analysis (3-panel)	sentiment_analysis.png	9.1

All figures saved to [reports/figures/](#) at 150 DPI.

11. Business Recommendations

Based on the full analysis, the following recommendations are made for different market participants:

For Hosts (Current Operators)

1. **Target Superhost status as a primary growth lever** — superhosts receive 3x more bookings at equivalent pricing. The return on service quality investment is booking volume, not higher prices.
2. **Price against active-market medians, not city averages** — the active-listing median (€213.68) is 20% higher than the naive median (€177.67). Hosts benchmarking against the wrong baseline are systematically underpricing.
3. **Prepare for 2028 now** — all HUT licenses expire November 2028. Hosts without a verified HUTB license (61.5% of listings by our analysis) face either compliance costs or forced exit from the market.

For Investors

1. **Example is the only neighbourhood tier that is statistically distinct from all others** — for premium positioning, it is the defensible choice. Upper-mid tier neighbourhoods (Sant Martí, Gràcia) are statistically indistinguishable from each other, offering competitive supply but no exclusive premium.
2. **Avoid interpreting the 2028 phase-out as an opportunity** — the supply reduction will not benefit remaining operators, as the legal mechanism eliminates the entire HUT license category, not just individual competitors.
3. **Revenue ceiling is capacity-driven, not location-driven** — the ML model shows [accommodates](#) (physical capacity) outweighs neighbourhood dummies in price prediction. Larger-capacity properties offer more pricing headroom than location premiums.

For Platform/Policy Analysts

1. **Market concentration warrants monitoring** — 0.2% of hosts control 17.6% of listings. The market structure is closer to institutional property management than peer-to-peer home-sharing.
2. **License compliance is a major gap** — only 890 verified HUTB numbers found across 15,293 listings (5.8%). Even accounting for multi-unit licenses and exemption categories, a large share of listings operate in a legally grey zone.
3. **Sentiment and numerical scores provide complementary signals** — a combined review-quality metric (NLP sentiment + structured scores) would more accurately surface listings with genuine guest goodwill vs. structurally inflated ratings.

12. Limitations & Caveats

1. **Single snapshot** — all findings reflect a single scrape (June 2026). No longitudinal tracking; trends inferred indirectly.
 2. **Calendar lacks pricing** — revenue analysis relies on `estimated_revenue_1365d` (Inside Airbnb pre-computed), not independently derived.
 3. **12 columns permanently null** — `host_since`, `host_response_time`, `instant_bookable` and 9 others are missing for all 15,293 listings in this scrape; host tenure and response analysis are unavailable.
 4. **VADER English-only** — 41.1% of reviews excluded from sentiment analysis. Multilingual coverage requires GPU-scale transformer models.
 5. **ML model biases** — systematic underprediction in Nou Barris (-15.5%) and shared rooms (-14.1%); model not suitable for pricing advice in these segments without correction.
 6. **License field is self-reported** — HUTB numbers are host-declared; validity/expiry status not independently verified.
 7. **Inside Airbnb methodology** — data is scraped, not from Airbnb's API. Prices may reflect listed rather than booked rates.
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13. Future Improvements

1. **Amenity text features** — parse the `amenities` JSON field into binary flags (pool, parking, balcony etc.) and include in ML model; estimated R^2 improvement of 0.02-0.05
 2. **Multilingual sentiment** — XLM-RoBERTa covers 100+ languages; would extend NLP analysis to full review corpus
 3. **Separate medium-term rental model** — `minimum_nights` dominates as a proxy for rental type; training separate models for short-stay vs. medium-term would improve both
 4. **Longitudinal scraping** — scrape the same city across multiple months to enable real trend analysis rather than snapshot inference
 5. **Listing description NLP** — topic modeling and keyword extraction on listing descriptions to identify quality/positioning signals
 6. **True RAG system** — build a retrieval-augmented generation system over the full reviews corpus for natural-language market querying
 7. **Production MLOps** — implement model retraining pipeline triggered by new scrape ingestion, with drift monitoring and automated alerts
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14. Reflection

Prioritization decisions

This submission deliberately chose depth over breadth. The decision to analyze a single city (Barcelona) rather than multiple cities was made early and held consistently — the assignment's own design philosophy states "quality outweighs quantity," and the regulatory context of Barcelona offered external richness that a multi-city surface-level approach would have sacrificed.

The most time-intensive phases were data engineering (Phase 3) and statistical analysis (Phase 5) — both high-weight rubric categories. The ML section (Phase 6) was scoped deliberately to 3 models with rigorous

validation rather than a larger experiment grid, prioritizing honest residual analysis over impressive-looking leaderboard numbers.

What I would do differently

- Build the demand segmentation view earlier (Phase 2 rather than discovering it mid-EDA) to avoid retroactive filtering across notebooks
- Include amenity features from the start of ML feature engineering
- Attempt multilingual sentiment analysis via a HuggingFace zero-shot classifier rather than defaulting to English-only VADER

Key lessons

- Real data is messier than assignment descriptions suggest — 12 columns being 100% null across an entire dataset is not in any textbook example
- Effect size is not optional — H5 demonstrated this conclusively with 5.58M observations producing $p < 10^{-16}$ for a 0.39pp difference
- Document decisions as you make them, not from memory — the assumptions log written in real-time was far more accurate and detailed than anything reconstructable retrospectively

Appendix A: AI Usage Disclosure

Tools Used

Tool	Version/Model	Purpose
Claude (Anthropic)	claude-sonnet-4-6	Project guidance, code review, debugging assistance, report structure
Groq API	llama-3.3-70b-versatile	LLM market intelligence briefing generation (Section 9.2)

AI-Assisted Sections

- **Code:** Pipeline scripts and notebook cells were developed interactively with Claude providing initial implementations which were then debugged, corrected, and validated against actual data outputs
- **Report structure:** Report outline and section framing developed with Claude; all numerical findings, interpretations, and business conclusions are the candidate's own analysis of real data
- **LLM briefings (Section 9.2):** Generated by Llama-3.3-70b via Groq; all input statistics are pre-computed from the actual dataset; LLM acted as writer not analyst

Key Prompts

Available on request — the primary interaction was a multi-turn guided analysis session where each phase was completed, verified with real data output, and corrected before proceeding. Full conversation available.

Output Validation

- Every code block was run against actual data and output verified before proceeding

- Statistical results cross-checked for direction and magnitude plausibility
- LLM briefing output verified against source findings for factual accuracy
- Multiple AI-suggested conclusions were rejected or modified when data contradicted them
(documented in assumptions log: Finding 12 geojson encoding, Finding 11 Barceloneta hypothesis)

Critical Assessment

Claude's initial hypothesis about the geojson encoding issue (Finding 12) was wrong — the garbling was a terminal display artifact, not a real data mismatch. This was caught by programmatic equality testing rather than accepting the visual output. Similarly, the initial Barceloneta "small subdivided apartments" hypothesis (Finding 11) was incorrect and corrected by the data. These corrections are documented in the assumptions log as examples of not accepting AI-generated hypotheses without verification.